Idrees Mustafa Aziz

Machine Learning Researcher

idrees.aziz@metu.edu.tr | Jammu & Kashmir, India
github.com/idreesaziz | linkedin.com/in/idrees-m-aziz

Research on LLM evaluation, learning from simulated data, and time-series prediction.

EDUCATION

Middle East Technical University (METU) | Ankara, Türkiye
B.E. in Computer Engineering | Expected 2027

Graduate-level Probability Theory *(in progress)*. Coursework: Machine Learning; Statistics for Computer Engineers; Linear Algebra; Calculus I-III; Scientific Computing; C Programming; Data Structures; Algorithms.

RESEARCH EXPERIENCE

Independent Machine Learning Researcher Spring 2026 - Present

No One Gets an A: Distributional Bias in LLM Text Scoring

Code

Research study | **First author** | 2026 | **Study manuscript**

- Designed a study of **11 LLMs and ~99,000 text assessments** to test whether 0-10 writing scores reflected deliberate reductions in text quality.
- Tested five corrections for scores that clustered despite differences in text quality. One improved how often **better text ranked above worse text from 51.8% to 64.9%** across nine models.
- Used **bootstrap confidence intervals** to check uncertainty and **mixed-effects models** to account for differences between the source articles.

Recovering Pre-Event Crowd Beliefs in Prediction Markets with a Risk-Calibrated Mixture of Inverse Experts

Preprint

7th ACM International Conference on AI in Finance (ICAIF 2026)
Accepted; selected for oral presentation

Sole author | Proceedings forthcoming, November 2026

- Built **APD-Safe** to recover a crowd's original probability estimate from trading records. Used **boosted trees** to combine estimates from several models of trading behaviour.
- On **1,500 simulated markets from ten types absent from training**, recovered beliefs with **12.8% less error** than averaging prices after each trader's first trade.
- Across **98 real markets**, recovered beliefs with **10.2% less error (RMSE)** than averaging the prices after each trader's first trade. Tested against pre-trading surveys from projects excluded from training.

RESEARCH CONTINUED

FinEntropyBench: What Does Entropy Tell Us About Market Efficiency? — Preprint

7th ACM International Conference on AI in Finance (ICAIF 2026)

Accepted; selected for oral presentation

Sole author | Proceedings forthcoming, November 2026

- Built a benchmark to test whether **entropy, a measure of randomness, reveals useful forecasting signals**. Tested an **LSTM** and linear and tree-based models across **16 market series**.
- Ran **113 tests** across individual markets and years, selecting models on earlier data and measuring their forecasts and trading returns in the following year.
- Patterns in the size of price changes did not reliably predict direction or trading profits.

Market Decompression: Learning Latent-State Inverse Maps for Cold-Start Financial Prediction — Preprint

7th ACM International Conference on AI in Finance (ICAIF 2026)

Accepted after peer review

Sole author | Not included in the proceedings

- Trained a **Transformer on simulated markets**, using hidden volatility and jump events as labels. Adapted it to forecast price variability with **five real training examples**.
- Used a **Gaussian-process model** to choose simulations likely to improve real forecasts. Compared identical models trained with and without those extra labels.
- Against this control, reduced **volatility forecast error (MSE) by 4.7% to 7.0% in three of five markets**. The other two failed to improve on later test data.

Central-bank communication graphs — Data/code

Sole author | Dataset manuscript under review at *Scientific Data*

- Built a public **ML dataset from 2,872 Bank of England and Federal Reserve documents**, with graphs of linked information and numerical text representations.
- Checked records for consistency and linked them to source documents so others could verify the data.

TECHNICAL SKILLS

Programming: Python, C++, SQL, TypeScript.
ML and analysis: PyTorch, NumPy, pandas, SciPy, scikit-learn; model training, simulation, controlled comparisons, and statistical evaluation.
Tools: Git, Linux, LaTeX.